PCA

Summary sheet — Course Name (Author / Instructor)

Core idea & notation

PCA is a purely *descriptive*, exploratory method (no probabilistic model assumed, normality not required) that handles any number of quantitative variables. It moves from p variables to $q < p$ new variables while distorting reality as little as possible.

- X $(n \times p)$ — data: n individuals (rows), p variables (columns); x_i^k = value of variable k for individual i.
- V $(p \times q)$ — projection matrix; $U = XV$ $(n \times q)$ — projected data.
- a_j — eigenvalue of axis v_j (= inertia carried by the axis); $X' = UV^T$ — reconstruction.

1. Goal and principle

Goal: find the projection that **minimizes information loss**, i.e. the one that **maximizes the dispersion (variance)** of the projected data. **Variance:** mean squared distance between the points and their centroid c: $\frac{1}{n}\sum_{i=1}^{n} d(x_i, c)^2$.

Property: going from p to $q < p$ necessarily loses dispersion, hence variance. The idea is to lose as little as possible by choosing the projection axes well.

Key principle of PCA. Choose the projection (the axes) so as to *maximize the variance* of the projected points.

2. The matrices X, V, U explained

Symbol	Size	Meaning
X	$n \times p$	original data; row i = individual $x_i = (x_i^1, \ldots, x_i^p)$; column k = variable X^k
v_j	$p \times 1$	j-th column of V = *weights* of the linear combination defining the new variable V_j (a direction in the original space)
V	$p \times q$	$V = [v_1\ v_2\ \cdots\ v_q]$: one column per new axis
U	$n \times q$	coordinates of the individuals in the new space; u_i^j = value of individual i on new variable j

New variables (linear combinations of the initial ones):

$$V_j = v_j^1 X^1 + v_j^2 X^2 + \cdots + v_j^p X^p, \qquad v_j = (v_j^1, \ldots, v_j^p), \qquad \|v_j\|^2 = \sum_k (v_j^k)^2 = 1$$

- V_1: the axis on which points are spread the most (↑ variance).
- V_2: must spread points as well, but *not correlated* to V_1 (V_2 brings new information); and so on.

Projection:

$$\boxed{U = XV} \qquad \text{entry by entry:} \quad u_i^j = \sum_{k=1}^{p} x_i^k\, v_j^k = \langle x_i, v_j \rangle$$

Reconstruction: $X' = U\,V^T$ $(n \times p)$. **Objective:** find V such that X' is as close as possible to X:

$$\min_{U,V}\ J(U,V) = \|X - UV^T\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{p} (x_{ij} - x'_{ij})^2$$

Case $q = 1$ (Eckart & Young, 1936): the unique solution v is the eigenvector of the largest eigenvalue of X^TX, with $\|v\|_2 = 1$.

3. Algorithm

1. Center and reduce X (normalization).
2. Covariance matrix: X^TX.

3. Eigenvalues and eigenvectors of $X^T X$.
4. Sort the pairs (a_i, v_i) by decreasing eigenvalue.
5. Keep the first q pairs: $V = [v_1 \ \cdots \ v_q]$.
6. a_i = inertia carried by axis v_i. Then $U = XV$.

	Centering only	**Centering + scaling**
Operation	subtract the mean of each variable	also divide by the standard deviation
Use when	variables are *homogeneous* (same meaning, unit, scale)	variables are *heterogeneous*

Outliers. PCA is very sensitive to noise: in the lecture's example a few outliers change the number of axes needed to reach the same explained variance (96 %).

4. Worked example with X, U, V (toy data, $n = 4$, $p = 2$)

Setup: four individuals, two variables, columns already centered (sums = 0) and of the same scale, so centering is enough:

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{pmatrix} \quad (4 \times 2)$$

1. **Center/reduce:** column means are 0 ⇒ nothing to subtract; same scale ⇒ no scaling.
2. **Covariance:** $X^T X = \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}$.
3. **Eigen-decomposition:** $\det(X^T X - aI) = (10 - a)^2 - 64 = 0 \Rightarrow a = 18$ or $a = 2$. $v_1 = \frac{1}{\sqrt{2}}\binom{1}{1}$ for $a_1 = 18$; $v_2 = \frac{1}{\sqrt{2}}\binom{1}{-1}$ for $a_2 = 2$ (both unit norm, orthogonal).
4. **Sort:** $(a_1, v_1) = (18,\ v_1)$ then $(a_2, v_2) = (2,\ v_2)$.
5. **Keep $q = 1$:** $V = v_1 = \frac{1}{\sqrt{2}}\binom{1}{1}$ $(p \times q = 2 \times 1)$.
6. **Project:** $U = XV$ (4×1); with all axes, $V = [v_1\ v_2]$ and U is 4×2:

$$U = XV = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & 1 \\ 3 & -1 \\ -3 & 1 \\ -3 & -1 \end{pmatrix}, \qquad \text{e.g. } u_1^1 = \tfrac{2 \cdot 1 + 1 \cdot 1}{\sqrt{2}} = \tfrac{3}{\sqrt{2}} \approx 2.12$$

Result: first column of U $(= u^1)$ is the $q = 1$ representation. Reconstruction with $q = 1$: $X' = u^1 v_1^T$ gives row 1 = (1.5, 1.5) instead of (2, 1); $J = \|X - X'\|_F^2 = 4 \times 0.5 = \boxed{2 = a_2}$, the eigenvalue of the discarded axis. Explained variance of $q = 1$: $\frac{18}{18+2} = \boxed{90\,\%}$.

5. Explained variance: choosing q

$$\text{choose the smallest } q \text{ such that } \quad \frac{\sum_{i=1}^{q} a_i}{\sum_{i=1}^{p} a_i} > s$$

s = chosen threshold. Since a_i is the inertia of axis i, this is the proportion of information kept.

Axis	1	2	3	4
Eigenvalue a_i (lecture, 4 grades)	28.23	12.03	0.03	0.01
Explained variance	0.70	0.30	0.00	0.00
Cumulative	0.70	1.00	1.00	1.00
Toy example above	18	2	—	—
Explained / cumulative	0.90	0.10 / 1.00	—	—

In the lecture's example ($\sum a_i = 40.30$), $q = 2$ keeps all the information.

6. Studying the variables: flip the table and use correlations

Idea: position yourself in a different space: the table is read *transposed* (each row = a variable, each column = an individual). Each initial variable x^i (column i of X) and each principal component c^j (column j of U) is then a vector of $\mathbb{R}^n$. **Goal:** analyze the linear correlation coefficients $r(c^j, x^i)$ between initial variables and principal components.

Procedure (same spirit as the PCA steps):

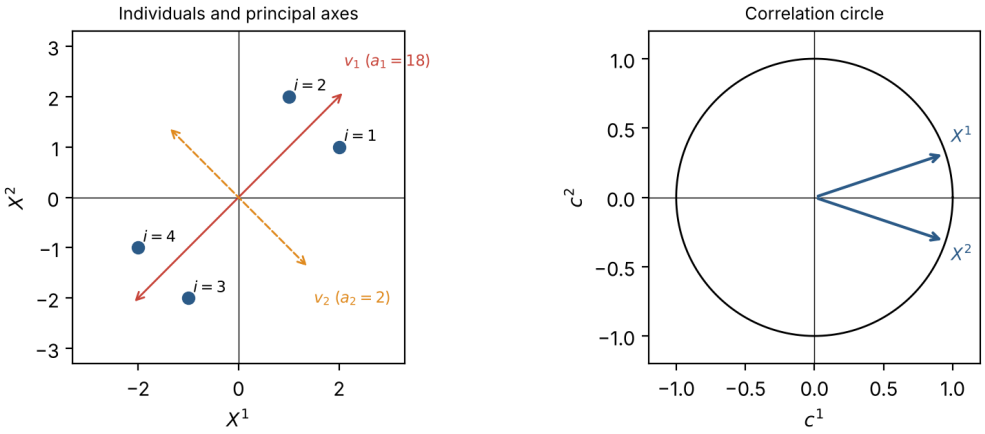
1. Compute the components: $U = XV$ (columns $c^j = u^j$).
2. For every variable i and axis j: $r(c^j, x^i) = \dfrac{\sum_k u_k^j \, x_k^i}{\sqrt{\sum_k (u_k^j)^2}\sqrt{\sum_k (x_k^i)^2}}$ (all columns are centered).
3. Gather them in a $p \times q$ table of correlations ("variables–factors").
4. Plot variable x^i as the point $\big(r(c^1, x^i),\, r(c^2, x^i)\big)$ inside the **correlation circle**.
5. Read: two arrows close to each other ⇒ correlation near 1; arrows at right angle ⇒ correlation near 0.

Example: toy data with $x^1 = (2, 1, -1, -2)$, $x^2 = (1, 2, -2, -1)$, $u^1 = \frac{1}{\sqrt{2}}(3, 3, -3, -3)$, $u^2 = \frac{1}{\sqrt{2}}(1, -1, 1, -1)$:

$$r(c^1, x^1) = \frac{18/\sqrt{2}}{\sqrt{18}\sqrt{10}} = \frac{3}{\sqrt{10}} \approx 0.95, \qquad r(c^2, x^1) = \frac{\sqrt{2}}{\sqrt{2}\sqrt{10}} = \frac{1}{\sqrt{10}} \approx 0.32$$

Variable	c^1 (90 %)	c^2 (10 %)
X^1	0.95	0.32
X^2	0.95	−0.32

Check: $0.95^2 + 0.32^2 = 1$ (arrow length = 1, on the circle, as all axes are kept). The two arrows are close: cos of their angle = $0.9 - 0.1 = 0.8$, which is exactly the correlation of x^1 and x^2 (8/10). *Shortcut:* since $X^T u^j = a_j v_j$, $r(c^j, x^i) = v_j^i \sqrt{a_j} / \|x^i\|$.



Toy example. Left: individuals and principal axes v_1, v_2. Right: correlation circle, X^1 and X^2 are strongly correlated.

Reading the circle. Long arrows near the circle are well represented by the kept axes; short arrows near the centre are poorly represented, so do not interpret them (see X^5, X^6 vs X^4 in the lecture).

7. Quick-reference checklist

Check	Tool	Pass looks like
Normalization	centering (+ scaling if heterogeneous)	column means $= 0$
Axes	eigenpairs (a_i, v_i) of $X^T X$, $\|v_i\| = 1$	sorted $a_1 \geq a_2 \geq \cdots$
Choice of q	$\sum_{i \leq q} a_i / \sum_{i \leq p} a_i$	above threshold s
Individuals	$U = XV$	similar points are close in the plane
Variables	$r(c^j, x^i)$, correlation circle	long arrows near the circle
Outliers	inspect the data first	no isolated points dominating axis 1

Exam reflex: (1) center (and scale) $X \to$ (2) $X^T X \to$ (3) eigenpairs (a_i, v_i), sorted $\to$ (4) pick q with $\sum_{i \leq q} a_i / \sum_{i \leq p} a_i > s \to$ (5) $V = [v_1 \cdots v_q]$, $U = XV \to$ (6) individuals in U, variables via $r(c^j, x^i)$ in the correlation circle.